

# MOHAMMAD ALI KHAN

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Software Developer and Cybersecurity graduate with 7+ years of experience developing enterprise banking applications and customer-facing financial systems.

Experienced in secure software development, web applications, backend systems, and database technologies using C#, .NET, Java, Python, C++, and SQL platforms. MSc in Computer Science for Cybersecurity from Oxford Brookes University and CompTIA Security+ certified.

Seeking opportunities in cybersecurity, SOC operations, application security, or secure software engineering.

## EDUCATION

### Oxford Brookes University

MSc Computer Science for Cybersecurity

2022 - 2023

Oxford, UK

#### Modules

- Malware analysis.
- Secure programming
- Operating Systems Security.
- Enterprise Networking.
- Secure Systems Architecture.

**Thesis:** "RSA- Really Staying Alive" Research project analyzing RSA implementation, security considerations, and comparison with Elliptic Curve Cryptography.

### National University of Singapore

BEng Electrical Engineering

2010 - 2014

Singapore

#### Modules

- Power electronics.
- Electric Drives and Control.
- Physics for Electrical Engineers.
- Signals and Systems.
- Programming Methodology.
- Programming for Computer Interfaces.
- Data Structures and Algorithms.
- Engineering Professionalism.

**Thesis:** "Optical meta-surface to display a 3-D image in air" investigating the optical properties of 'metasurfaces' and how they alter the path of rays of light.

## WORK EXPERIENCE

### R-ECO (Renewable Energy Cooperative)

Heat Pump Heating System Designer

2024 - 2026

Oxford, UK

- Bespoke design of all Heat Pump Heating Systems within the scope of MCS (Microgeneration Certification Scheme) regulations.
- Design of heat emitters as part of the heating system.
- Heat Pump System performance estimates, Heat Loss Reports, Emitter Design Reports and Heat Pump Quotes.
- Working with sales operations teams to ensure that systems are designed according to client needs.

### Freelance work

Freelance jobs via Upwork

2023 - 2024

- Software development and programming jobs:
  - Tweaked old code to work with newer versions of C++ for a company based in India
  - Scripts to pull specific data from webpages for a small firm in India
- Python, Bash scripting and PowerShell related work.

**Bank AL Habib Limited**  
*IT Officer Grade I (Programmer/Software Developer)*

**2015 - 2022**  
*Karachi, Pakistan*

- Developed and maintained internal banking applications used by operational staff across multiple departments, improving document handling and workflow management.
- Contributed to digital banking and financing systems supporting customer onboarding and loan processing operations.
- Development and maintenance of a Web banking application, using ORACLE Service Oriented Architecture and ORACLE Banking Digital Experience (OBDX).
- Software and web development work in MVC.NET C#, JAVA, Database work in IBM DB2, Microsoft SQL Server and MySQL.
- Interviewing and onboarding new team members.

## **COURSES AND CERTIFICATIONS**

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**CompTIA** **2025**

*SY0-701 - CompTIA Security+ certification exam*

- A certification that showcases professionals' capabilities in securing networks, applications, and devices, ensuring data integrity, confidentiality, and availability.

**Udemy** **2019**

*Machine Learning A-Z™: Hands-on Python & R in Data Science*

- Covered an array of topics in Machine Learning, including Dimensionality Reduction, Reinforcement Learning, and Deep Learning, using Python and R programming languages for instruction.

**MITx** **2018**

*Introduction to Computational Thinking and Data Science*

- Course on Data Science concepts, Inferential Statistics, and Dynamic programming techniques.

*Introduction to Computer Science and Programming Using Python*

- An introductory course on Computer Science and Programming; Covered Computational thinking and algorithms, and Python programming practices.